

Especificação de Interfaces do Sistema

Smart Decisions

1. Especificação de APIs (Swagger / OpenAPI - YAML)

A seguir estão definidos os principais endpoints de cada microserviço, em formato compatível com OpenAPI 3.0 (Swagger).

1.1 API Gateway

```
openapi: 3.0.0
paths:
  /canvas/{id}/analyze:
    post:
      summary: Solicita análise do Canvas
      security:
        - bearerAuth: []
      parameters:
        - in: path
          name: id
          required: true
          schema:
            type: string
      responses:
        '202':
          description: Análise iniciada
        '400':
          description: Canvas inválido
```

1.2 Canvas Service

```
paths:
  /canvas:
    post:
      summary: Criar novo Canvas
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Canvas'
      responses:
        '201':
          description: Canvas criado

  /canvas/{id}:
    get:
      summary: Obter Canvas por ID
      parameters:
        - in: path
          name: id
          required: true
```

```

        schema:
          type: string
      responses:
        '200':
          description: Canvas retornado

/canvas/{id}/analysis/latest:
  get:
    summary: Obter última análise do Canvas
    responses:
      '200':
        description: Análise retornada

```

1.3 Intelligence Service

```

paths:
  /intelligence/analyze:
    post:
      summary: Processa análise inteligente do Canvas
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/CanvasData'
      responses:
        '200':
          description: Análise processada
        '500':
          description: Falha na LLM

```

1.4 Market Data Service

```

paths:
  /market-data/sector:
    post:
      summary: Obter dados de mercado por setor
      description: Consulta múltiplas APIs externas e retorna dados consolidados do setor informado.
      requestBody:
        required: true
        content:
          application/json:
            schema:
              type: object
              properties:
                sector:
                  type: string
                  example: "SaaS"
                region:
                  type: string
                  example: "Brasil"
      responses:
        '200':

```

```
    description: Dados de mercado retornados
  '503':
    description: Falha ao consultar APIs externas
```

1.5 Auth Service

```
paths:
  /auth/validate:
    post:
      summary: Validar token JWT
      description: Verifica autenticidade, expiração e escopos do token.
      requestBody:
        required: true
        content:
          application/json:
            schema:
              type: object
              properties:
                token:
                  type: string
      responses:
        '200':
          description: Token válido
        '401':
          description: Token inválido ou expirado
```

1.6 Notification Service

```
paths:
  /notifications/send:
    post:
      summary: Enviar notificação ao usuário
      requestBody:
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/Notification'
      responses:
        '200':
          description: Notificação enviada
```

2. Fluxo de Eventos (Mensageria Assíncrona)

O sistema utiliza mensageria para desacoplamento entre serviços e processamento assíncrono da análise inteligente.

2.1 Evento: CanvasAnalysisRequested

```
{
  "eventType": "CanvasAnalysisRequested",
  "timestamp": "2026-01-16T12:00:00Z",
  "canvasId": "uuid",
  "userId": "uuid",
  "canvasData": {
```

```
    "segments": "...",
    "valueProposition": "...",
    "channels": "...",
  }
}
```

Publicado por: Canvas Service

Consumido por: Intelligence Service

2.2 Evento: CanvasAnalysisCompleted

```
{
  "eventType": "CanvasAnalysisCompleted",
  "timestamp": "2026-01-16T12:05:00Z",
  "canvasId": "uuid",
  "analysisId": "uuid",
  "score": 82
}
```

Publicado por: Intelligence Service

Consumido por: Notification Service e Canvas Service

2.3 Evento: NotificationDispatch

```
{
  "eventType": "NotificationDispatch",
  "userId": "uuid",
  "message": "Sua análise de negócio está disponível.",
  "channel": "EMAIL"
}
```

Publicado por: Notification Service

Consumido por: Provedor de Email/Push

2.4 Evento Opcional: MarketDataFetchFailed

```
{
  "eventType": "MarketDataFetchFailed",
  "timestamp": "2026-01-16T12:03:00Z",
  "sector": "SaaS",
  "reason": "Timeout in external API"
}
```

Publicado por: MarketDataService

Consumido por: Intelligence Service (fallback/log)